

CHAPTER 5

Material Handling, Distribution, and Routing

5.1. Introduction

This chapter discusses the management of *material movement* in an organization. Material movement occurs anytime a product or part thereof moves or is transported from one place to another. Material movement usually accounts for somewhere between 5 and 90% of total factory cost, with a probable average of 25%. Since little or no work is done on a part or product as it moves, movement in that sense is nonproductive; yet, on the average, a factory spends 25% of its dollars on this movement.

Material movement occurs throughout the entire product manufacturing cycle and both before and after it. Raw materials are usually moved from the natural state to some type of primary operation before finally being moved to the manufacturing plant. After production, the product is moved or distributed to the various users. After the product has completed its useful life, it is either discarded or scrapped. For discarding, one more movement occurs before the material is disposed; but in the case of scrappage, movement occurs back to the primary operation for reclamation, thus completing what might be called a material movement cycle. This can be represented as shown in Figure 5.1. Notice that movement occurs throughout the entire cycle. For example, not only does movement occur between phases, but it also occurs within a phase (e.g., material handling within the plant). This diagram

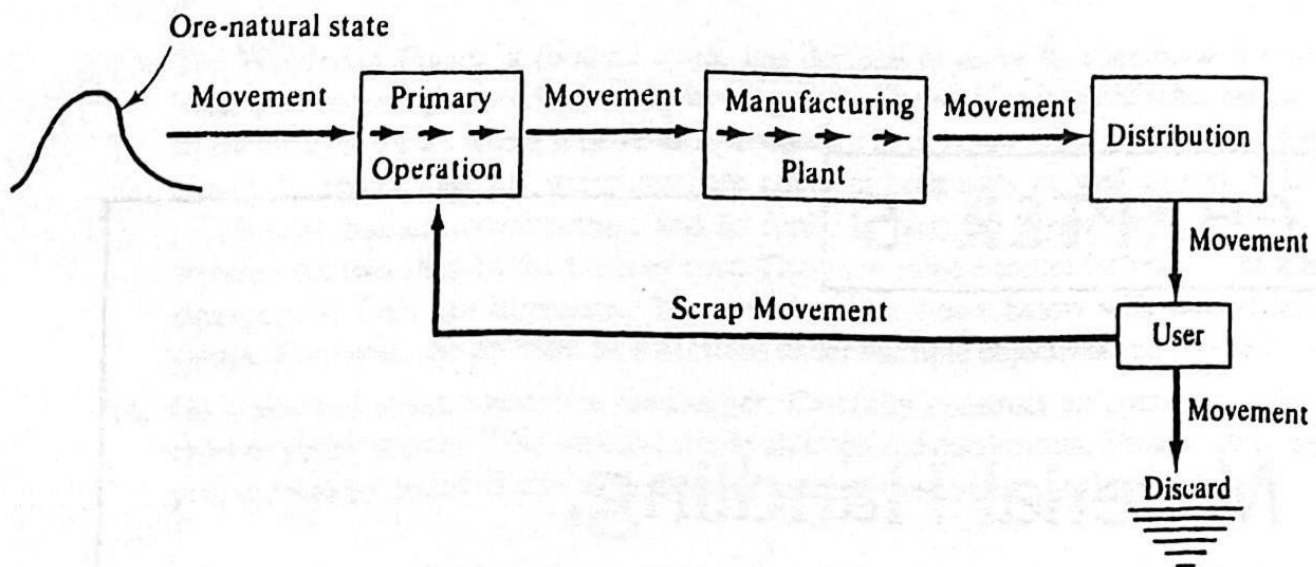


Fig. 5.1. Material movement cycle.

was simplified by showing distribution as one box when actually it could involve several boxes (e.g., wholesaler, retailer, warehouses, regional warehouses, etc.).

Movement is, of course, not restricted to the private sector of the economy. School buses carrying children to and from schools are an exceptionally important material movement; and so are street sweepers, mailmen, and bookmobiles. All these are examples of movement which require energy and therefore should be managed well.

5.2. Material Handling

In this section we present a short discussion on *material handling*. Although material handling is handling any material, any place, and any time, we will limit our discussion here to within-plant movement and leave distribution (outside-plant movement) to be covered in the next section. Before beginning, however, let us go to Corvallis, Oregon, to see how material handling affects Gadgets, Inc.¹

Example 5.1

Being an astute manager who is very cost conscious, you have always been aware of where your money goes and have tried to cut expenses whenever possible. Not until you made a study of material movement, however, did you realize just how important this area was. You find that movement occurs in getting all your raw materials and purchased parts to the plant and within the plant to the appropriate areas. Within the plant, you might find that material movement accounts for approximately 25% of total factory cost and that little or no productive work is involved in movement. Finally, you might find that distribution costs

¹In future chapters, we will visit Gadgets, Inc. frequently to see how what we are studying affects the plant. More details on the plant will be given as needed.

are very high and that substantial savings are available through good industrial engineering management of distribution.

After becoming cognizant of all the above, you would probably immediately initiate a study of material handling methods. With the subject matter presented in this chapter, you are able to conduct a detailed thorough study and come up with improved material handling effectiveness at less cost.

5.2.1 Equipment Concepts

The first problem encountered in material handling is the almost impossible task of understanding what types of equipment are available and what are the strengths and weaknesses of each. There are available more than 500 different types and varieties of material handling equipment, with each material handling system usually requiring several different types. This coupled with the rapidly changing technology makes it very difficult to keep up to date in material handling. Some of the basic types of material handling equipment are listed below.²

- (1) *Conveyors*—Gravity or powered devices used for conveying material from one fixed point to another fixed point (although portable conveyors may be moved from time to time). They are particularly useful for moving homogeneous materials over a route that does not vary and when the material is supplied at a relatively constant rate. Figure 5.2 is an example of an extendible belt conveyor useful for loading and unloading trucks.
- (2) *Industrial Trucks*—Vehicles (either manual or powered) used for transporting material over varying paths where usually the load is intermittent. They are especially useful when the nature of the load and the path prohibit the use of conveyors or other mass production oriented devices. Job shops, for example, find industrial trucks particularly useful. Figure 5.3 is a picture of a powered high-lift electric fork truck.
- (3) *Cranes and Hoists*—Overhead lifting devices used for moving loads in a fixed area. They are particularly useful for moving intermittent loads when the size or shape varies between loads. They, of course, require some ceiling clearance and adequate support from the roof or building columns. Figure 5.4 is a picture of a coil lifter suspended from an overhead crane.
- (4) *Containers and Racks*—Devices used to store and handle bulk materials. Containers are used to unitize quantities of material for storage or use. Racks are storage devices for better use of space (higher utilization of the cube). Figure 5.5 shows a sample storage rack application and Figure 5.6 is a picture of a plywood container being stacked for storage purposes.
- (5) *Elevators and Lifts*—Devices used to vertically raise or lower material. Although fork trucks, conveyors, cranes, and hoists also lift, these devices lift

² Adapted from James M. Apple, *Lesson Guide Outline on Material Handling Education*, College Industry Committee on Material Handling Education, Pittsburgh, Pa., May 1975, p. 68.

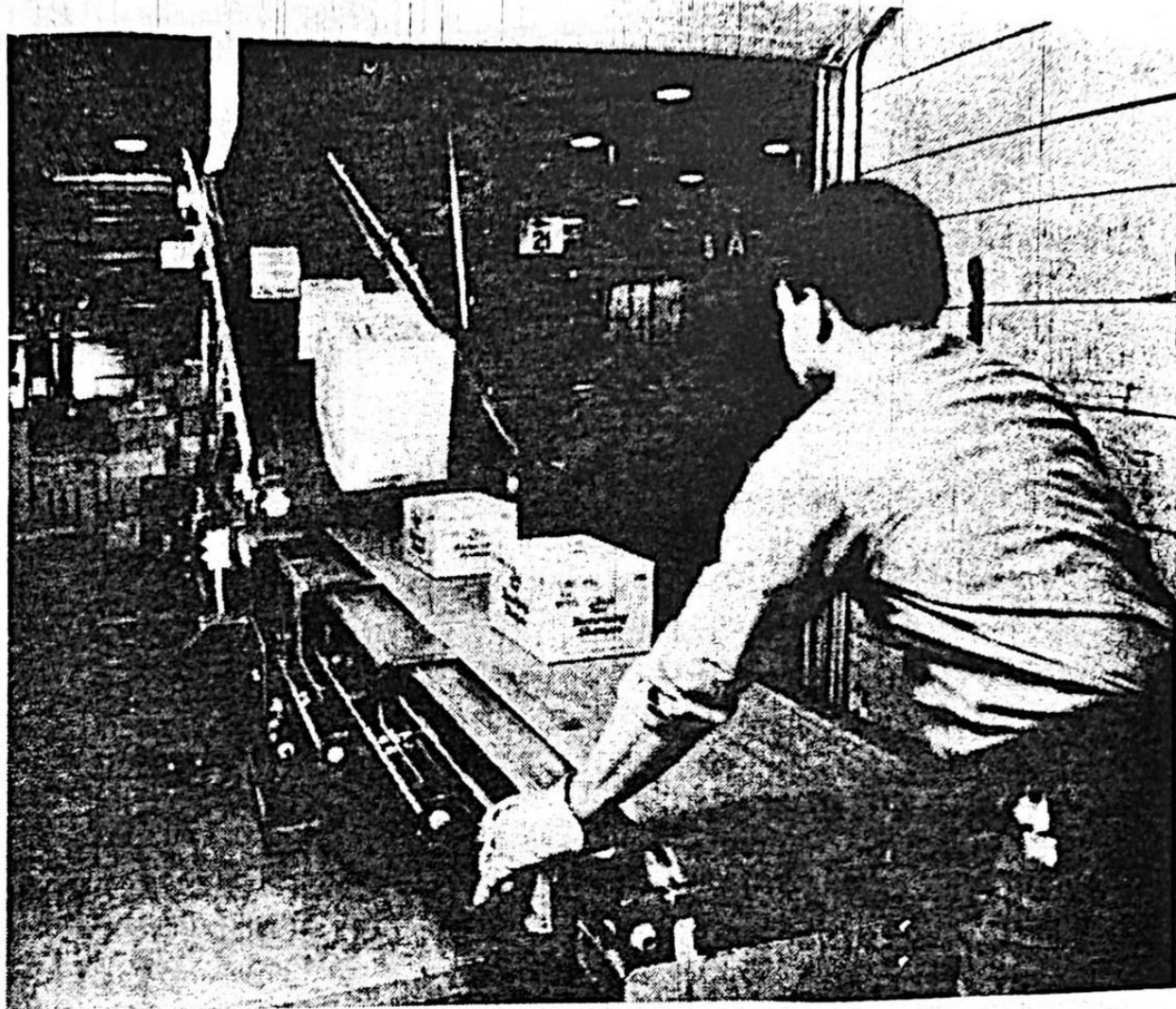


Fig. 5.2. Extendible belt conveyor.



Fig. 5.3. High-lift electric fork truck.

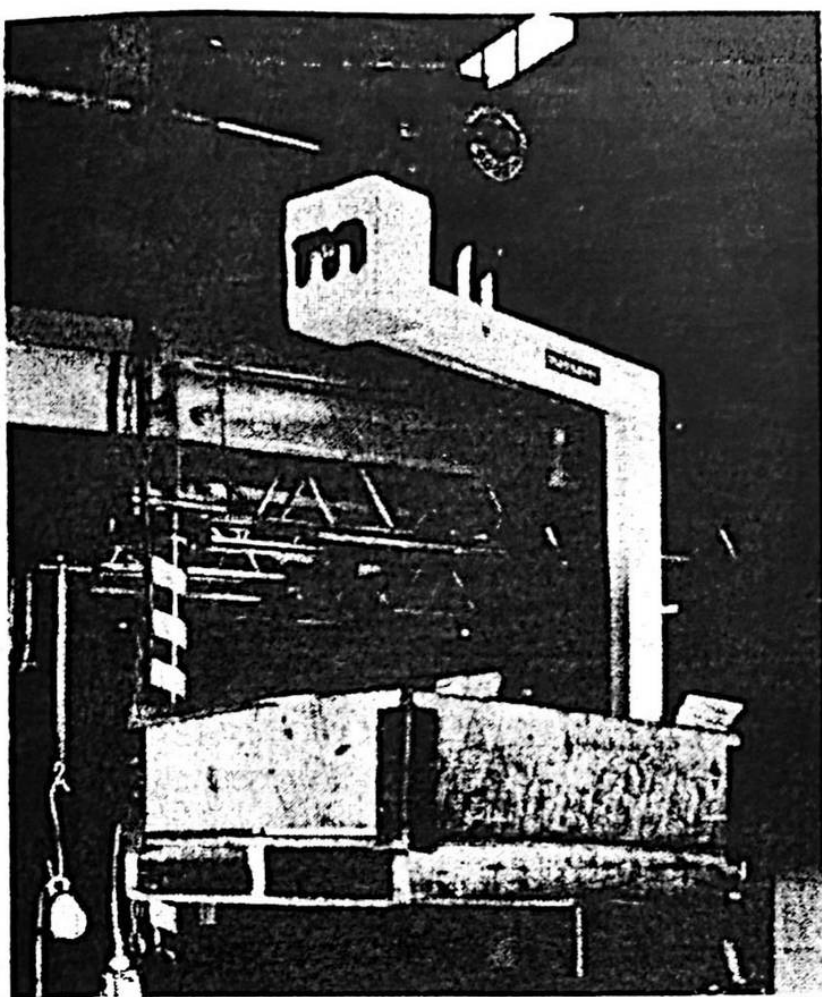


Fig. 5.4. Coil lifter suspended from overhead frame.

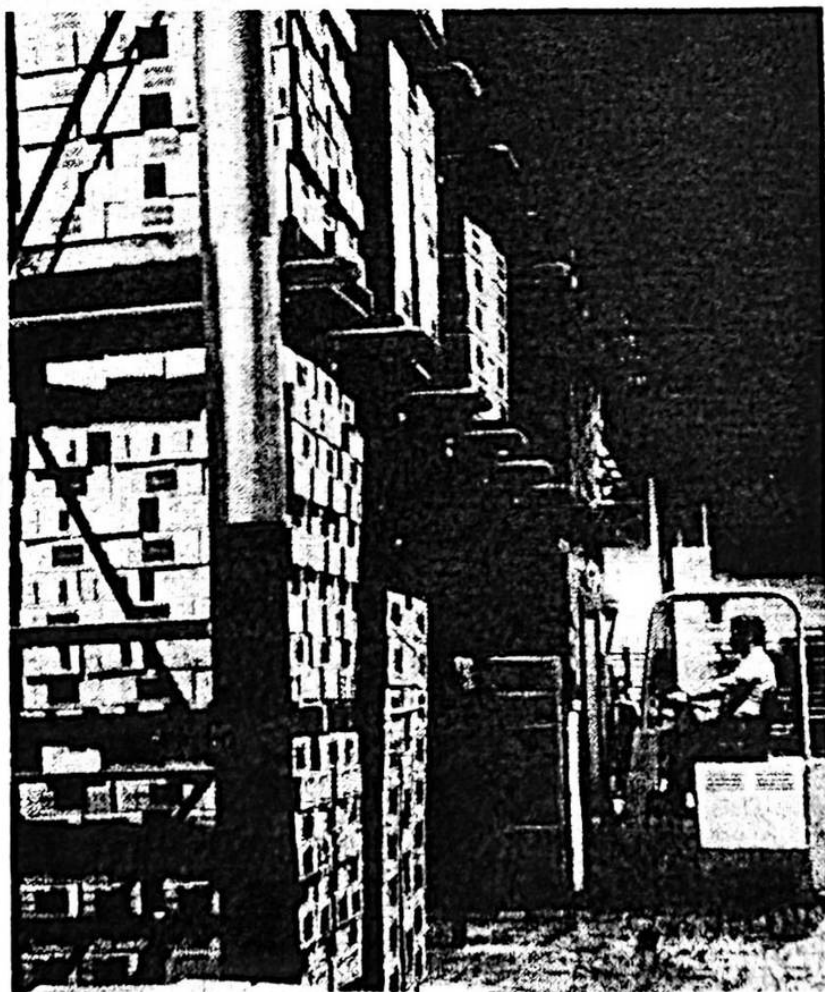


Fig. 5.5. Storage rack application.

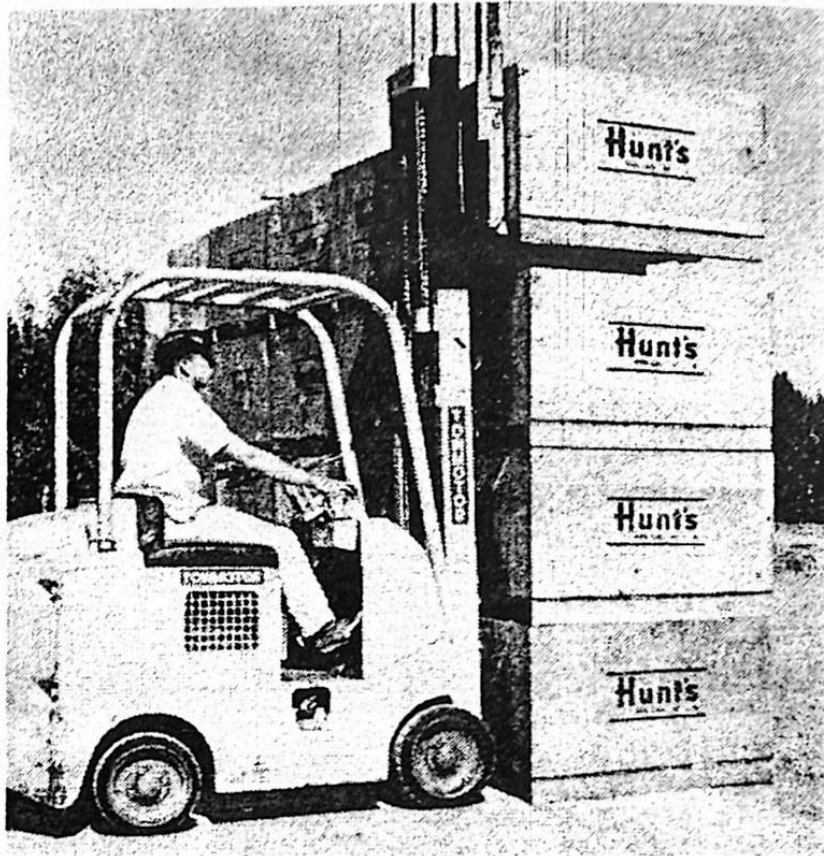


Fig. 5.6. Stacked plywood pallets.

by applying force from underneath and are fixed in location. Figure 5.7 is a picture of a hydraulic scissor lift.

- (6) *Automatic Guided Vehicles (AGVs)*—Driverless vehicles capable of moving along predetermined paths and performing certain prescribed duties. Usually, AGVs are relatively nonintelligent in that they can only go where they are

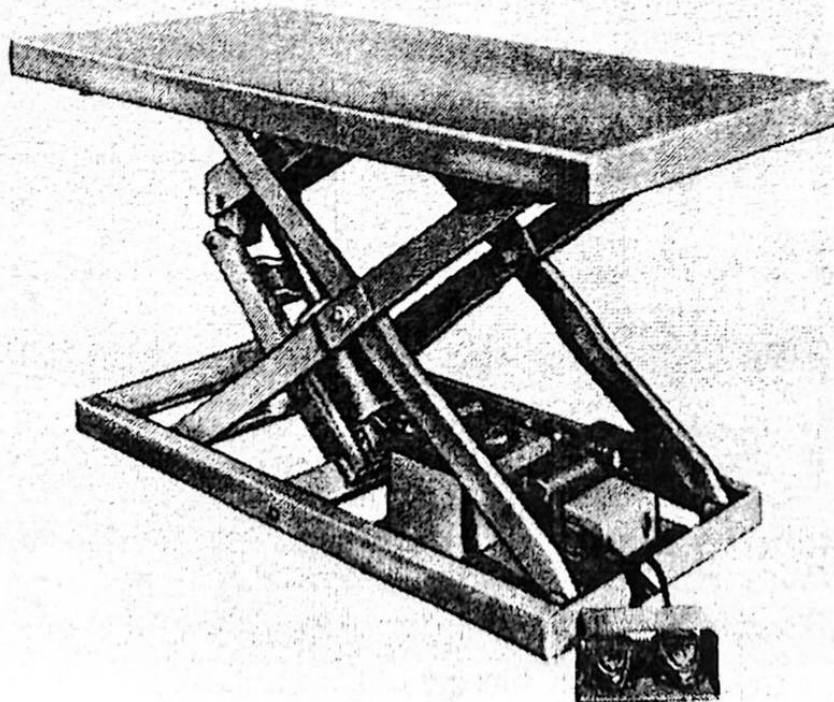


Fig. 5.7. Hydraulic scissor lift.

sent and perform a very limited activity, such as "stop at point A." Figure 5.8 is a picture of an AGV.

- (7) *Automatic Storage and Retrieval System (ASRS)*—Storage rack systems where storage locations and inventory records are kept by the control system. When product is needed, the computer automatically finds the storage location and a driverless (usually, crane) system automatically retrieves the product. The reverse happens for products going to storage. These systems are very popular today and are useful in reducing labor costs and in achieving high-density storage. Figure 5.9 is a picture of an ASRS in a flooring manufacturing operation.

The industrial engineer must have a working knowledge of each of these basic types of material handling equipment. With this knowledge, he or she is able to make decisions on the equipment needed and can proceed to do detailed economic analysis.

5.2.2 Principles of Material Handling

Given that you now understand the basic types of material handling equipment, you must still design systems that effectively and efficiently utilize the equipment. Over the years certain principles of material handling that lead to efficient and effective systems have been observed and recorded in list form for use by those involved in material handling systems design.³

- (1) *Planning Principle*—All material handling systems should be planned, not evolved.
- (2) *Systems Principle*—The planning of the system should include as much as possible of the material movement cycle. For example, a vendor could ship material in a container that can be used throughout your production process and perhaps even used as the container for shipment to a customer.
- (3) *Material Flow Principle*—Facilities layout and process engineering should assure that materials flow smoothly.
- (4) *Simplification Principle*—Apply the concepts of work simplification to material handling systems design. For example, the principles of motion economy should be obeyed.
- (5) *Gravity Principle*—Gravity should be used to move material whenever practical. Gravity feed bins and roller conveyors are examples.
- (6) *Space Utilization Principle*—Make optimum use of space. High racks for use with narrow aisle stacking trucks make good total "cube" usage.
- (7) *Unit Size Principle*—In general, the larger the accumulated load handled, the better. The use of pallets to handle large quantities of properly stacked cartons is an example. Unitization should occur as soon as possible and as long as

³Extracted, with permission, from James M. Apple, *Material Handling Systems Design* (New York: Ronald Press, 1972).

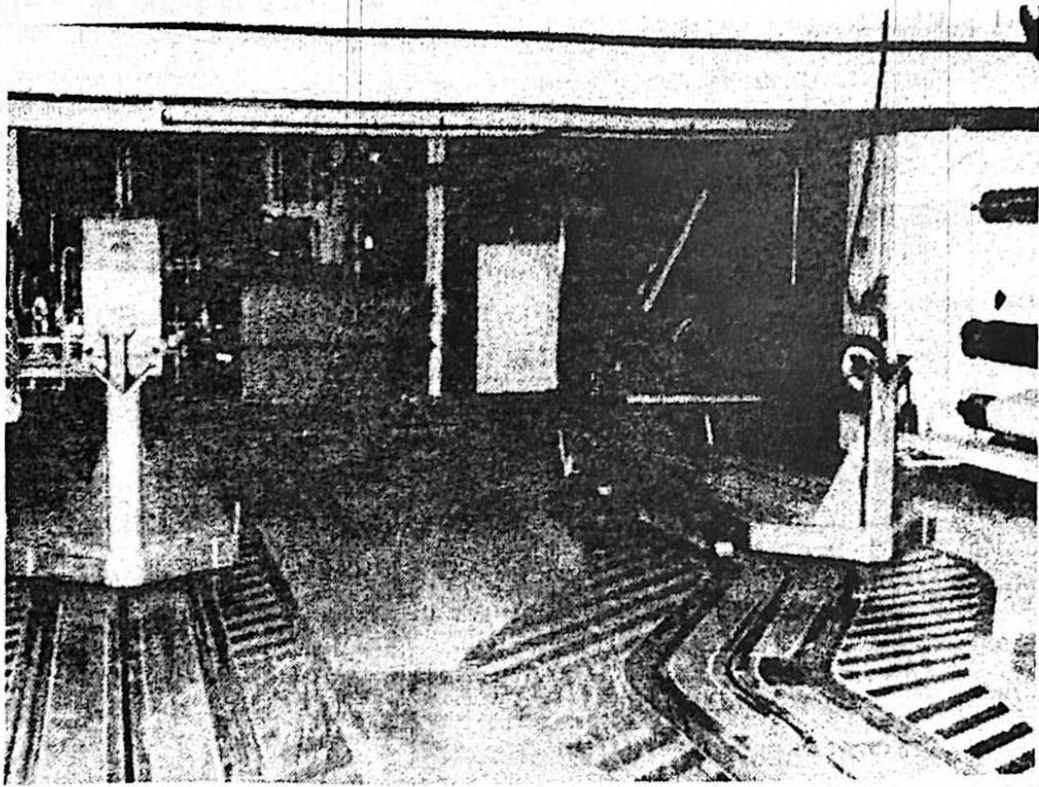


Fig. 5.8. Automatic guided vehicle.

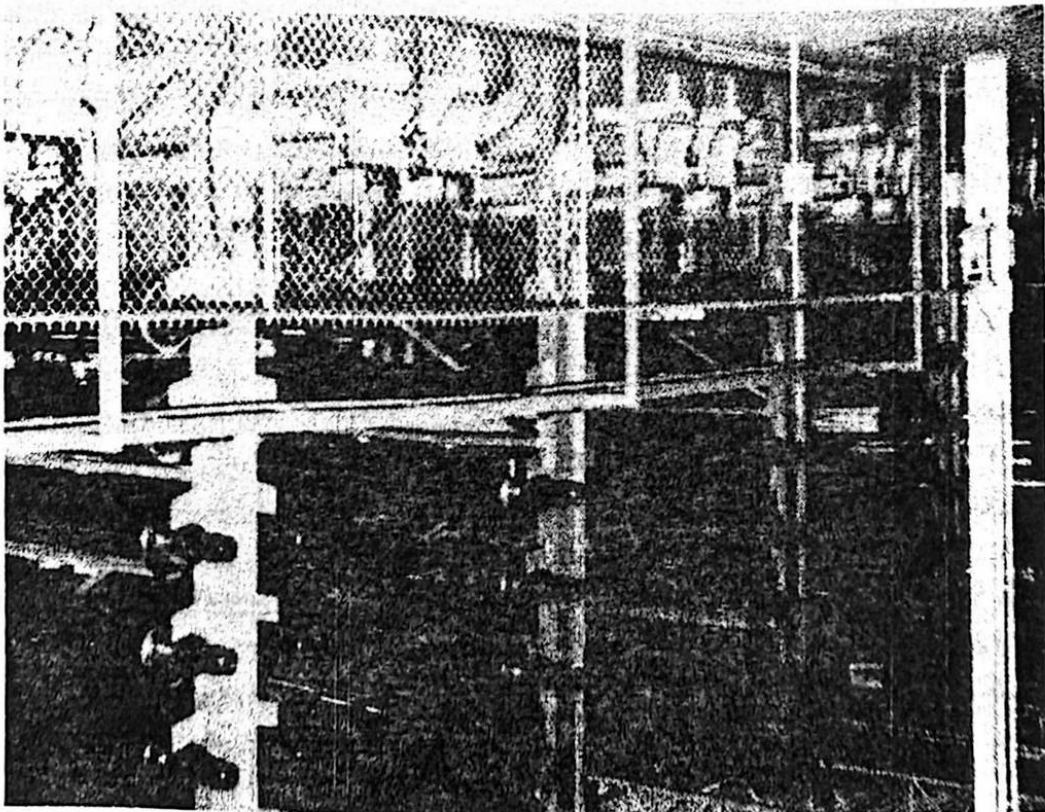


Fig. 5.9. Automatic storage retrieval system.

possible in the cycle and stacking patterns should be developed for more effective unitization.

- (8) *Mechanization and Automation Principle*—Always consider mechanization and automation in the design of the handling system. They may be impractical but they should be considered. Powered conveyors are an example of mechanization and automatic pallet stackers is an example of automation.
- (9) *Equipment Selection Principle*—Be careful to consider all aspects of the material to be handled, the move to be made and the methods to be utilized, in selecting equipment. The purchase of fork trucks with a great deal of flexibility for use throughout the plant is an example.
- (10) *Standardization Principle*—Standardize material handling methods and equipment. The adoption of a system of pallet stacking patterns and the use of standard-sized pallets are examples.
- (11) *Adaptability Principle*—Design methods and equipment so they can perform a variety of tasks. Variable-speed conveyors are examples.
- (12) *Dead Weight Principle*—Reduce the ratio of equipment or dead weight to payload. The use of plastic, aluminum, or other lightweight pallets is an example.
- (13) *Utilization Principle*—Effectively utilize both manpower and equipment. The use of walkie talkies by fork truck drivers for rapid assignment to new jobs is an example.
- (14) *Maintenance Principle*—Plan for preventive as well as emergency maintenance on material handling equipment. The use of a preventive maintenance program and the adequate stocking of replacement parts are examples.
- (15) *Obsolescence Principle*—Make economic evaluations periodically to check on replacement of material handling equipment by identical models or newer types. Scheduled replacement policy based on sound engineering economy is an example.
- (16) *Control Principle*—Use material handling equipment to aid production control, inventory control, quality control, etc. The use of conveyors for assembly lines to ensure a constant rate is an example.
- (17) *Capacity Principle*—Use handling equipment to aid in fully utilizing production capacity. The use of outdoor storage whenever possible and the design of conveyor speeds for optimum feeding of parts are examples.
- (18) *Performance Principle*—Measure and monitor the performance of your material handling system. For example, the calculation of a cost per move and the constant monitoring of this should be done.
- (19) *Safety Principle*—Design methods and equipment for operator safety. The use of “cages” around and especially over the head of a fork truck operator is an example.

These 19 principles are vitally important to anyone involved in material handling systems design and each principle should be considered in every design endeavor. They are, however, qualitative in nature. Almost all industrial engineers

prefer to deal with quantitative techniques. In fact, many quantitative techniques are applicable in the design of material handling systems, as we shall see in the next section.

* 5.2.3 Quantitative Techniques

In Section 4.3.1 we presented a technique called *transportation programming* that was used in the location of facilities. That same technique can be used in material handling systems design. As an example, a transportation problem can be formulated for the allocation of fork trucks, electric walkie trucks, pallets, and hand trucks to various sites throughout the plant where they are needed. Another technique very similar to transportation programming is called the *assignment technique*. This technique allocates scarce resources just as transportation programming does, but it can only handle one unit at a time.

For example, suppose that you have a fork truck located at each of five different sites throughout the plant, but these sites no longer have need for the trucks. Suppose, further, that there are five additional sites that need fork trucks. Since the distances between the sites vary substantially, you want to assign the fork trucks to the five new sites and minimize the total distance traveled. Let us assume that the data are as given in Figure 5.10.

Since your objective is to assign trucks from existing sites to sites that need trucks, the problem is similar to a transportation problem. Only one truck is available or needed at each site, however; therefore, the assignment problem is simpler than the transportation problem, and it is more easily solved. One alternative solution to the problem (*not optimal*) is given in Figure 5.11.

Notice that this solution has *one and only one* entry in each row and in each

		Locations Needing Trucks				
		Shipping (A)	Inspection (B)	Maintenance (C)	Receiving (D)	Dock Storage (E)
Present Locations of Trucks	From \ To					
	Tool Room (A)	10	5	7	3	2
	Press Shop (B)	9	4	8	6	5
	Paint Storage (C)	2	1	4	3	2
	Welding (D)	8	7	9	10	6
	Assembly (E)	3	2	5	8	1

Entries in matrix are distances in hundreds (00's) of feet.

Fig. 5.10. Assignment formulation.

	A	B	C	D	E
A				1	
B			1		
C					1
D		1			
E	1				

Assign Fork Truck From A to D
Assign Fork Truck From B to C
etc.

Fig. 5.11. Typical solution to assignment problem.

column. Going back to the original distance matrix, we can subtract a number from every element in a specific row or column without affecting the solution (the value of the solution will be changed by the amount of the number subtracted; but not the solution itself). For example, since the smallest number in the first row is 2, subtract 2 from every entry in that row. This yields

8 3 5 1 0

The smallest number in the second row is 4; subtracting 4 from every entry in that row and repeating for all other rows yields Figure 5.12. Now, if this can be done for rows, it can also be done for columns, as shown in Figure 5.13.

The reduced matrix B is of interest to us for we can now solve the assignment problem and be certain that the solution is a solution to the original one (by our observation that subtracting a number from every entry in any row or column does not affect the solution). Since there are no negative numbers in the reduced matrix B, an assignment that has zero total cost must be optimal. We can see by inspection that assigning the truck from the Tool Room (A) to Receiving (D), the truck from the Press Shop (B) to Inspection (B), the truck from Paint Storage (C) to Shipping (A), and so on, as shown in Figure 5.14, results in a zero cost solution.

This is the optimum solution to our problem, and its cost should be equal to the sum of the numbers subtracted (again by our earlier observation) or

$$\underbrace{2 + 4 + 1 + 6 + 1}_{\text{Rows}} + \underbrace{1 + 0 + 3 + 1 + 0}_{\text{Columns}} = 19$$

	A	B	C	D	E	Smallest number in row		A	B	C	D	E
A	10	5	7	3	2	2	A	8	3	5	1	0
B	9	4	8	6	5	4	B	5	0	4	2	1
C	2	1	4	3	2	1	C	1	0	3	2	1
D	8	7	9	10	6	6	D	2	1	3	4	0
E	3	2	5	8	1	1	E	2	1	4	7	0

Original matrix

Reduced matrix A

Fig. 5.12. Row reduction of assignment matrix.

Reduced Matrix A						Reduced Matrix B					
	A	B	C	D	E		A	B	C	D	E
A	8	3	5	1	0	A	7	3	2	0	0
B	5	0	4	2	1	B	4	0	1	1	1
C	1	0	3	2	1	C	0	0	0	1	1
D	2	1	3	4	0	D	1	1	0	3	0
E	2	1	4	7	0	E	1	1	1	6	0
	1	0	3	1	0						

Smallest number in column

Fig. 5.13. Column reduction of assignment matrix.

Going back to the original matrix, we find that solution is

$$\begin{array}{cccccc}
 3 & + & 4 & + & 2 & + & 9 & + & 1 & = & 19 \\
 \text{A-D} & & \text{B-B} & & \text{C-A} & & \text{D-C} & & \text{E-E} & &
 \end{array}$$

which agrees with the solution just obtained.

This procedure is actually the first phase of the *Hungarian method* which will be presented in detail in Chapter 14. More is necessary if a zero cost assignment cannot be found by inspection of the row and column reduced matrix. The first phase is repeated below:

- (1) Subtract the smallest number in each row from every entry in that row.
- (2) In the matrix resulting from step 1 subtract the smallest number in each column from every entry in that column.
- (3) In the matrix resulting from step 2 a zero cost assignment (if it can be made) is optimal. If not, complete the Hungarian method.

These two techniques are, of course, not the only quantitative techniques that can be used in material handling systems design. There are many others, for example, queueing theory, integer programming, linear programming, simulation, and transshipment programming. Many of these will be discussed in the chapters on operations research (Chapters 14 through 17).

Cost matrix						Solution					
	A	B	C	D	E		A	B	C	D	E
A	7	3	2	0	0	A				1	
B	4	0	1	1	1	B		1			
C	0	0	0	1	1	C	1				
D	1	1	0	3	0	D			1		
E	1	1	1	6	0	E					1

Fig. 5.14. Zero cost solution to reduced assignment problem.

5.3. Distribution

Distribution covers all aspects of delivering the final product to the customers. Actually, it is a part of material handling since we defined material handling as handling material any time and any place. For ease of presentation, however, we have chosen to cover distribution separately. A revisit to Corvallis, Oregon, should help set the stage.

Example 5.2

As president of Gadgets, Inc., you are vitally concerned with the marketing activities of your plant. Marketing is a large field in itself covering many aspects of daily operations. One of the biggest areas, however, is in maintaining a good distribution function so that all customers can be assured of good service. This entails rapid delivery of your product and economical operations.

One of the first problems you encounter is that of warehouse location. You have a stated policy of being able to supply every customer within a maximum of 48 hours after receipt of order. Using the material presented in this chapter and Chapter 4, you locate your warehouses accordingly.

Another problem immediately confronts you. How should you deliver the product to the warehouses and from the warehouses to the customers? This necessitates a feasibility study for maintaining your own fleet of vehicles. You conduct this study examining such areas as number and types of vehicles, replacement policies for the vehicles, and maintenance systems for the vehicles. Using the material covered in this chapter and other chapters in this book, you are able to make these studies and corresponding decisions. You are then ready to move into the next area. (To be continued.)

5.3.1 Warehouse Location

In Chapter 4 we discussed several problems of facilities location. For Gadgets, Inc., there is now the problem of locating warehouses such that any customer can be serviced within 48 hours of receipt of order. Since we can correlate time and distance very easily, the objective is similar to minimizing the maximum distance. Since we are also interested in efficient operations, the cost of warehouses is important. Solving a problem like this by using strictly analytical techniques is impossible because the problem is so complex.

Many people turn to *simulation* to solve problems such as this. We shall cover simulation later but we can introduce it here. Simulation is using statistical data on location (of customers in this case) and demand to "try out" various numbers and locations of warehouses. Using the speed of digital computers, we can rapidly simulate many different solutions and management policies and monitor the results. From these results, intelligent decisions can be made.

5.3.2 Operations Management—Routing

Once the warehouses have been located, it is necessary to operate the logistics system. This includes maintaining adequate inventories of finished products and

supplying warehouses and customers in an efficient and effective manner. Basic inventory models are developed in Chapter 7, many of which are applicable to this problem, but more advanced models are often required in multilevel systems such as the one we are now discussing. These advanced inventory models are beyond the scope of this text.

We can, however, discuss the delivery question in some depth. Let us assume that we have located the warehouses and have designed our inventory system. We are then concerned with the delivery of goods to warehouses and customers. Questions encountered here include such items as commercial vs. in-house fleet systems, size and composition of in-house fleets, and routing of vehicles to efficiently deliver goods. Except for the routing question, all of these can be handled by using the techniques of engineering economy for decision making among alternatives. We shall discuss routing in some depth.

Since Gadgets, Inc., is small, we can assume that one vehicle can supply all warehouses. This vehicle must start at the main plant, deliver super gadgets to each warehouse, and return to the main plant. There may be several trips involved, for example, visiting warehouses 1, 2, and 5 on the first route and warehouses 3, 4, and 6 on another, and so on, but for simplicity we can assume just one trip.

This is the famous traveling salesman problem which has received considerable research attention in industrial engineering and operations research. In this problem a salesman is located in one city and must visit several other cities. He must start in this one city, travel to all other cities, and return to the starting city while minimizing the total distance traveled.

There are many procedures for solving traveling salesman problems, some of which guarantee optimum solutions and some of which are heuristic. Almost all traveling salesman problems are solved heuristically and yield near optimum answers.

Perhaps the simplest of these heuristics is the "closest unvisited city" approach:

- (1) Start at any city and visit the closest unvisited city. Continue in this manner until all cities have been visited.
- (2) Repeat step 1 for all cities as a starting point. Choose the best of the resulting solutions.

As an example of this formulation and approach, consider the problem of supplying gadgets from Corvallis, Oregon. Let us suppose that there are six warehouses located in Stillwater, Oklahoma; College Park, Pennsylvania; Cleveland, Ohio; Bozeman, Montana; Tampa, Florida; and San Diego, California. Further, assume that one truck can supply all warehouses in one trip. The objective is to leave Corvallis and visit each of the six warehouses before returning to Corvallis while traveling a minimum distance. Since the truck driver has supplied us with the mileages (given in Figure 5.15), we are ready to start the solution procedure.

First, however, let's make some observations on this distance matrix. The truck driver has five friends scattered all around the country. Therefore, the mileages

From \ To	A	B	C	D	E	F	G	
(A) Corvallis	∞	14	21	20	6	24	9	Entries Are in Hundreds (00's of Miles)
(B) Stillwater	14	∞	10	9	9	10	11	
(C) College Park	21	10	∞	1	15	9	21	
(D) Cleveland	20	9	1	∞	14	9	20	
(E) Bozeman	6	9	15	14	∞	19	9	
(F) Tampa	24	10	9	9	19	∞	21	
(G) San Diego	9	11	21	20	9	21	∞	

Fig. 5.15. Travel distances for Gadgets, Inc.

given above are only approximate and correspond to the routes he would take. The distance matrix has ∞ 's down the main diagonal of the matrix, signifying that the truck cannot go from a city back to the same city. The distance matrix is symmetric; that is, the mileage from city i to city j is the same as the mileage from city j to city i . Finally, using our heuristic procedure, there will be seven routes (one for each starting point) and we will choose the best of these seven.

If we start with Corvallis, the closest city is Bozeman (E)—600 miles; the closest unvisited city to Bozeman is either Stillwater (B) or San Diego (G) (we will have to try both). Assuming that Stillwater is selected, the next closest is Cleveland (D)—900 miles. Continuing, we obtain the following:

City	A	E	B	D	C	F	G	A
Mileage	6	9	9	1	9	21	9	= 64

Going back and trying San Diego in place of Stillwater, we obtain

City	A	E	G	B	D	C	F	A
Mileage	6	9	11	9	1	9	24	= 69

Using city B as a starting point, we obtain

City	B	D	C	F	E	A	G	B
Mileage	9	1	9	19	6	9	11	= 64

and (a tie occurred at the first step)

City	B	E	A	G	D	C	F	B
Mileage	9	6	9	20	1	9	10	= 64

Notice that these are feasible tours even though it looks like we start at B. Actually, the tour (for the second one) would start at A and go A-G-D-C-F-B-E-A. (It makes no difference where one starts on a circle.)

Using the rest of the cities as starting points, we obtain

City	C	D	B	E	A	G	F	C
Mileage	1	9	9	6	9	21	9	= 64

City	C	D	F	B	E	A	G	C
Mileage	1	9	10	9	6	9	21	= 65

City	D - C - F - B - E - A - G - D							
Mileage	1	9	0	9	6	9	20	= 64
City	E - B - D - C - F - G - A - E							
Mileage	9	9	1	9	21	9	6	= 64
City	E - G - A - B - D - C - F - E							
Mileage	9	9	14	9	1	9	19	= 70
City	F - C - D - B - E - A - G - F							
Mileage	9	1	9	9	6	9	21	= 64
City	F - D - C - B - E - A - G - F							
Mileage	9	1	10	9	6	9	21	= 64
City	G - A - E - B - D - C - F - G							
Mileage	9	6	9	9	1	9	21	= 64
City	G - E - A - B - D - C - F - G							
Mileage	9	6	14	9	1	9	21	= 69

The best of these has a distance of 64 (6,400 miles) and is generated by several routes. This would quickly lead us to conclude that this distance must be optimal and is generated by several routes. Actually, the optimal is 60 which is generated by the route following:

Corvallis (A)–San Diego (G)–Stillwater (B)–Tampa (F)–College Park (C)–Cleveland (D)–Bozeman (E)–Corvallis (A)

This route is shown in Figure 5.16, and of course, the route could be run in reverse order.

Two observations can be made here. First, the heuristic solution procedure did not give us the optimal answer, but it did give us some good starting solutions and helped us examine the problem. As always, the analytical technique is only an aid to the decision process, not a substitute for it.

Second, almost all traveling salesman routing problems can be solved manually and will yield very good solutions if the stops are shown on a map or some physical display. Figure 5.16 dramatically demonstrates that the solution is obvious. Of course, since in almost all cases the optimum solution is not so obvious, analytical procedures could be a great help. Nevertheless, all routing problems should be displayed on a map so that solutions can be seen.

Now, all this works very well for a traveling salesman problem, but what happens when more than one truck is required? That is, what if two or more trucks are available to make the circuit? First, if two or more trucks are available but one could handle the entire load (capacity is no problem), then the formulation is called a *multiple traveling salesmen problem*. There are solution procedures designed to handle multiple traveling salesmen problems, but like the traveling salesman problem, there are no efficient ways of obtaining the exact solution.

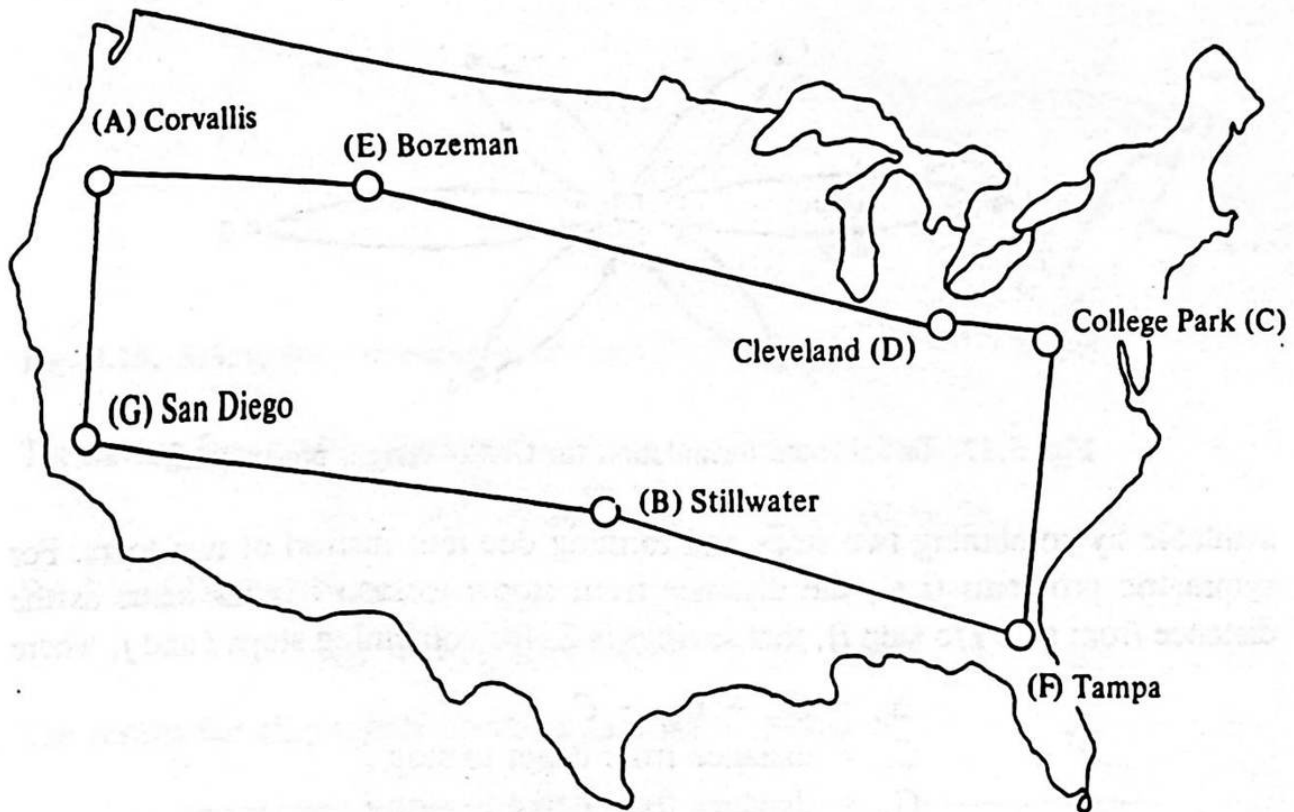


Fig. 5.16. Traveling salesman solution.

Usually, however, if there are two trucks, it means that one truck cannot handle the entire load; therefore, two trucks are required even if the solution with two trucks is more costly than that with one truck. This is also true for problems requiring more than two trucks.

This problem is called a *transportation routing problem* and is very similar to a *multiple traveling salesmen problem* except that there are capacity constraints on the vehicles. There are many solution procedures available for this problem, but, as with the traveling salesman problems, there are no efficient and effective techniques available today. In fact, very few transportation routing problems can be solved optimally because of the computational time and storage requirements on computers. Therefore, almost all problems are solved by using a heuristic procedure and there are many procedures available. One of the oldest, probably the most used, and perhaps conceptually the simplest is that developed by Clark and Wright.⁴

This procedure starts with the unrealistic assumption that each of the N stops or cities is serviced by a separate vehicle that starts from home base or the depot, goes to the stop and performs the service (e.g., delivers the goods), and returns to home base. Figure 5.17 depicts this situation. (It is important to realize that even though this assumption is made at the beginning, few, if any, final solutions will have such a situation.)

The next step in the Clark–Wright procedure is to calculate the savings

⁴G. Clark and J. Wright, "Scheduling of Vehicles from a Central Depot to a Number of Delivery Points," *Operations Research*, 12 (1964), 568.

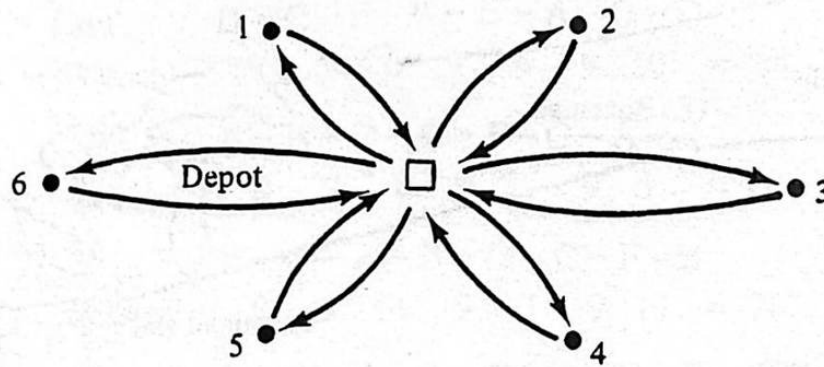


Fig. 5.17. Initial route formulation for Clark-Wright procedure.

available by combining two stops and forming one tour instead of two tours. For symmetric problems (i.e., the distance from stop i to stop j is the same as the distance from stop j to stop i), that savings is S_{ij} for combining stops i and j , where

$$S_{ij} = C_{oi} + C_{oj} - C_{ij}$$

C_{oi} = distance from depot to stop i

C_{oj} = distance from depot to stop j

C_{ij} = distance from stop i to stop j

This is demonstrated in Figure 5.18.

The Clark-Wright procedure then orders or ranks the savings in a list of decreasing magnitude such that the first on the list is the combination that yields the greatest saving, the second is the combination that yields the second greatest saving, and so forth on down the list. The procedure starts with the first combination on the list and makes the two stops into one route (if the constraints allow that combination) and continues down the list until the solution is complete.

As an example of this situation, consider your problem of Gadgets, Inc., as shown in Figure 5.16. This time, however, let us assume that there is enough demand on the warehouses so that at least two trucks are required. That demand is given in Table 5.1. Assume that each truck can carry 25,000 gadgets. Since total demand is for 42,000 gadgets, at least two trucks (or two visits by the same truck) are required.

The first step is to calculate the savings and rank them. That is done below. Notice that in this formulation the depot (Corvallis) has been removed so that there are only six stops.

From \ To	B	C	D	E	F	G
B	∞	10	9	9	10	11
C	10	∞	1	15	9	21
D	9	1	∞	14	9	20
E	9	15	14	∞	19	9
F	10	9	9	19	∞	21
G	11	21	20	9	21	∞

From \ To	B	C	D	E	F	G
A	14	21	20	6	24	9

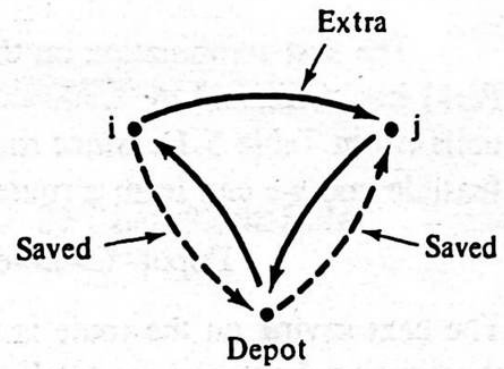


Fig. 5.18. Saving for combining stops i and j .

The saving for combining B and C is

$$AB + AC - BC = 14 + 21 - 10 = 25$$

For B and D, the saving is

$$AB + AD - BD = 14 + 20 - 9 = 25$$

The results for all possible combinations are as follows:

<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>
BC	25	CD	40	DF	35
BD	25	CE	12	DG	9
BE	11	CF	36	EF	11
BF	28	CG	9	EG	6
BG	12	DE	12	FG	12

Reranking these yields

<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>	<i>Stops</i>	<i>Saving</i>
1 CD	40	6 BD	25	11 BE	11
2 CF	36	7 BG	12	12 EF	11
3 DF	35	8 CE	12	13 CG	9
4 BF	28	9 DE	12	14 DG	9
5 BC	25	10 FG	12	15 EG	6

Table 5.1. DEMAND FOR GADGETS

	<i>Stop</i>	<i>Demand (gadgets)</i>
G	San Diego	10,000
B	Stillwater	5,000
F	Tampa	6,000
C	College Park	7,000
D	Cleveland	10,000
E	Bozeman	4,000

The first combination on the list is CD with a saving of 40. Stop C (College Park) has a demand of 7,000 units; stop D (Cleveland) has a demand of 10,000 units (from Table 5.1). Since the truck can hold 25,000 units, this combination is feasible and we can form a route consisting of

Depot-C-D-depot or depot-D-C-depot

The next saving on the route is for stops C-F. This would add another stop F to the existing route and the total demand would be

$$17,000 + 6,000 = 23,000 \text{ units}$$

Since the truck can handle this load, we now have one route consisting of

Depot-F-C-D-depot or depot-D-C-F-depot

The next saving on the list is for stops D and F. Since both D and F are already on the same route, the link is infeasible.

The next saving is for stops B and F. This would add another stop to the existing route and would create a total demand of

$$23,000 + 5,000 = 28,000 \text{ units}$$

Since the truck can handle only 25,000 units, this combination is infeasible.

The next combination on the list is for stops B and C, but since stop C is in the *middle* of the existing route, this link is infeasible. The next combination is also infeasible because it would combine stops B and D and would create a total demand of 28,000 units.

Next, stops B and G can be combined for a saving of 1,200 miles. Since the total load is only 15,000 units, we can create a new route that yields the following two routes:

Depot-F-C-D-depot or vice versa

and

Depot-B-G-depot or vice versa

The next two combinations (C and E; D and E) are both infeasible. The third is for stops F and G. Since this would combine the above two routes and would yield a total demand exceeding the truck capacity, this combination is also infeasible.

Then, stops B and E can be combined for a saving of 1,100 miles. This would add a third stop to the second route and would create a total demand of

$$15,000 + 4,000 = 19,000 \text{ units}$$

which is feasible.

Since this connects all stops (no combination of two routes is feasible), we can stop with the two routes being

Depot-F-C-D-depot or vice versa

and

Depot-E-B-G-depot or vice versa

The total distance traveled is

$$24 + 9 + 1 + 20 + 6 + 9 + 11 + 9 = 89 \text{ or } 8,900 \text{ miles}$$

Route 1 Route 2

The associated routes are shown in Figure 5.19.

Naturally, more distance is traveled than in the case of the traveling salesman problem because now there are two trucks. Since two trucks are required, the solution shown in Figure 5.19 is about the best that can be done, but the Clark-Wright procedure does not guarantee that the solution is optimal.

5.3.3 Routing in the Public Sector

Recently, industrial engineers have begun to solve more problems in public-sector activities. School bus routing is one area that has and still is receiving a great deal of attention. The average school bus costs well in excess of \$2.00 per mile to run exclusive of labor cost. Therefore, any savings in distance traveled would be helpful.

The same techniques used for private sector routing of vehicles can also be used for public sector routing. Particularly, the Clark-Wright approach has been used for school bus routing.

Another approach has been developed for use for smaller or more rural school districts that may not have access to computers.⁵ The procedure works this way:

- (1) Obtain a large map of the school district.
- (2) Plot stops and school locations with straight pins.
- (3) Starting at any convenient spot, make a sweep about the schools so that the area around an individual group of schools can be divided into wedges, each of which represents one bus route (Figure 5.20). "Close" each wedge as soon as enough students to fill one bus have been identified.
- (4) In each wedge, start with the bus stop farthest from the school and determine whether transferring that stop to a neighboring wedge can yield any saving in distance. If, for example, stop 4 is in wedge E, route the buses in wedge D or A so that they pick up at that stop. Calculate the distance saved by placing stop 4 in D or A and include the stop in whichever wedge gives minimum distance. During this "interchanging" keep a close count on the number of students involved. Although the buses in a wedge can exceed capacity during interchanging, make sure that numbers are leveled off before drawing final routes.

⁵J. A. Robbins and W. C. Turner, "No Computer? Try Wedging Bus Routes," *Nation's Schools and Colleges*, May 1975, pp. 44-45.

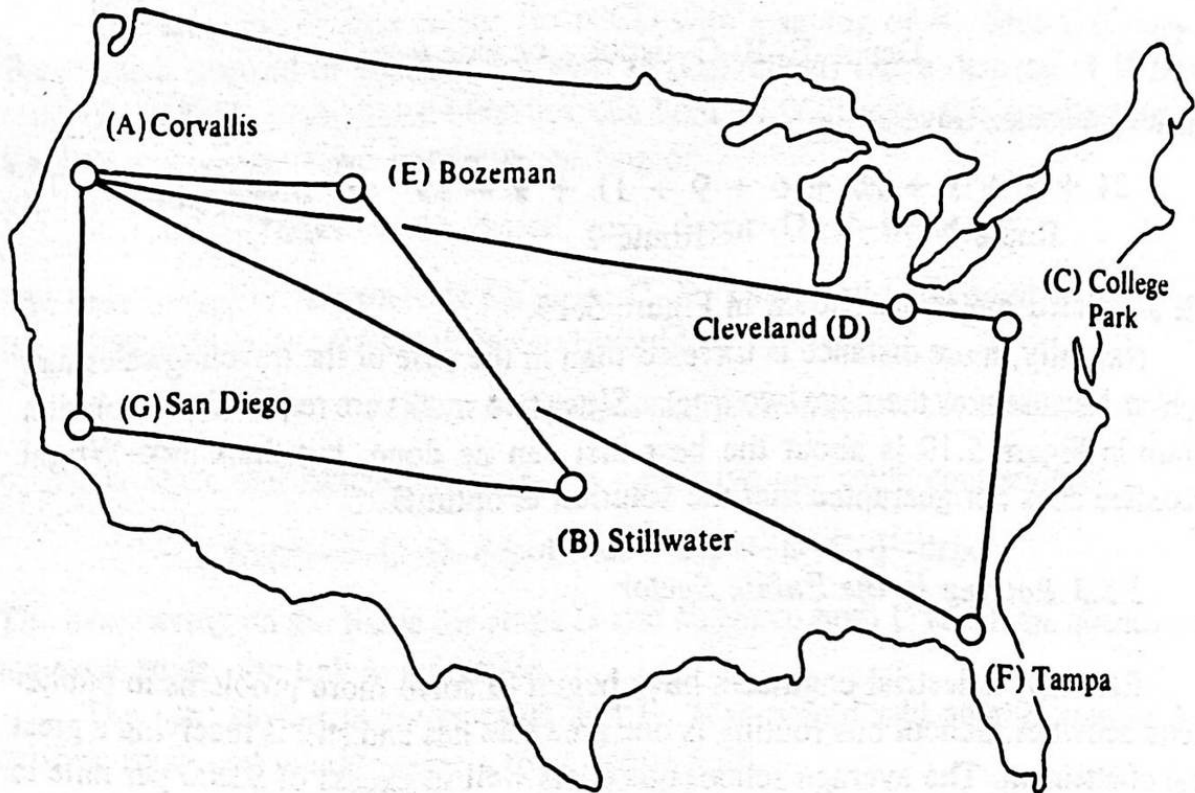


Fig. 5.19. Transportation routing solution.

- (5) Once a stop has been interchanged, go through the procedure again, starting with the farthest stop left in a particular wedge. It may now be beneficial to interchange stops that were not feasible to transfer before—because interchanging stops in any two wedges may have a domino effect on the entire system.
- (6) When it's no longer advantageous to interchange, begin with the stop farthest from the school in each wedge and route the buses into the schools (Figure 5.21).

School buses in a sample rural district with two elementary schools and one high school were rerouted by this method. Before the study was made 14 buses traveling 445 miles per day served the schools.

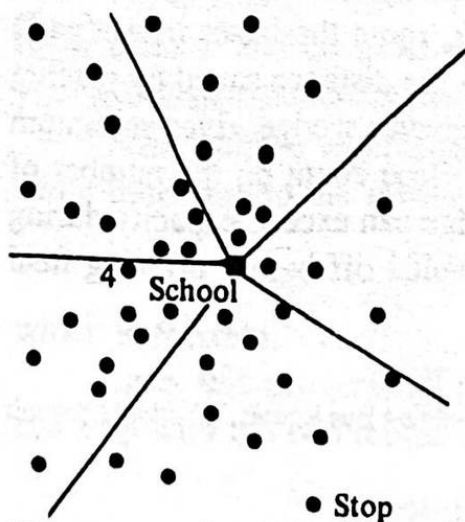


Fig. 5.20. Wedging for school bus routing.

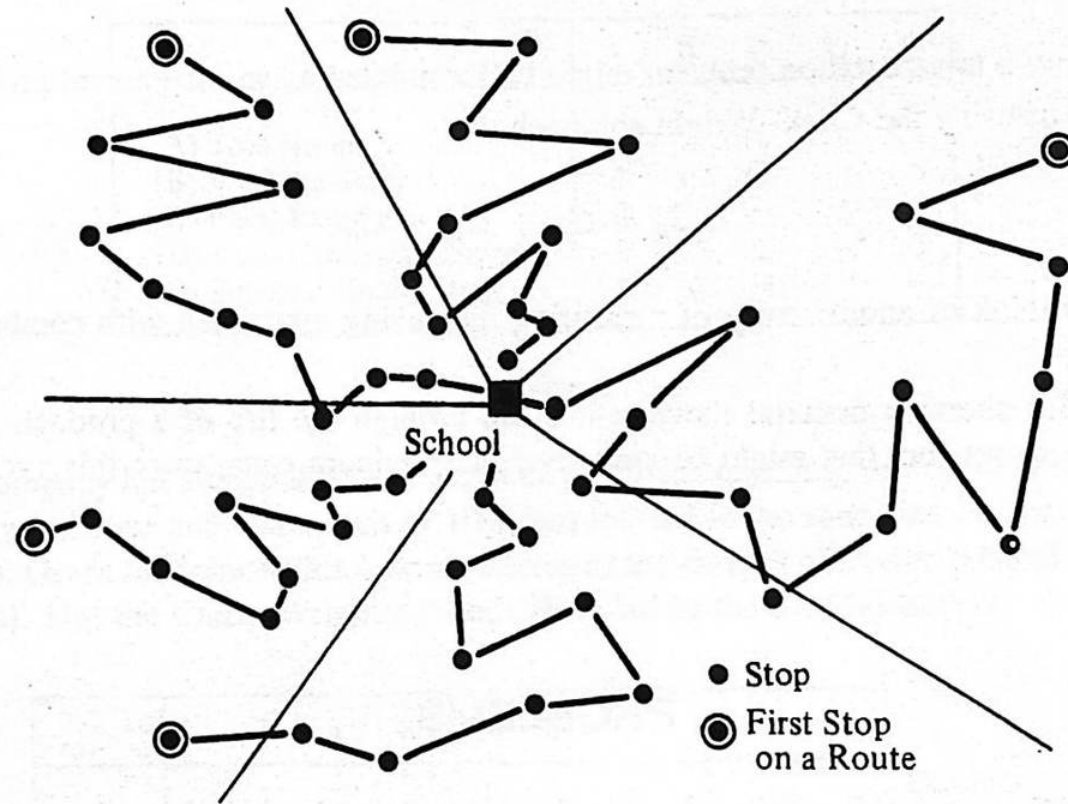


Fig. 5.21. Routing with wedges.

When the simplified procedure described was used, it was determined that 14 buses were indeed necessary, but that savings could be made in distance traveled. The new routing resulted in a total daily distance of 377 miles—a saving of 15.3%, or 68 miles per day. Maximum mileage on any single route did not change substantially, and students did not have to ride buses any longer than they did before. In another rural district the same procedure saved approximately 10% of the total mileage formerly used.

This area of routing in the public sector is a problem of growing concern because of the increased emphasis on productivity and energy conservation. It is attracting a great deal of attention in many industrial engineering programs.